

COMPARISSON OF TIME COMPLEXITY FOR LINEAR AND BINARY SEARCH

Feature	Linear Search	Binary Search
Method	Checks elements one by one from the beginning	Repeatedly divides the search space into two halves
Data Requirement	Works on both sorted and unsorted data	Requires sorted data
Best Case Time Complexity	$O(1)$	$O(1)$
Average Case Time Complexity	$O(n)$	$O(\log n)$
Worst Case Time Complexity	$O(n)$	$O(\log n)$
Implementation	Simple	Slightly more complex
Suitable for	Small datasets or unsorted data	Large sorted datasets

Which Algorithm is best suitable for search function:

Binary Search.

Because

- i. Search faster
- ii. Improve Performance
- iii. Scalability
- iv. Reduced time complexity